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Dielectric Properties and Insulation Performance of Nanocellulose Composite Films Enhanced by Hyperbranched Boron Nitride

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ABSTRACT

Boron nitride (BN), characterised by its wide band gap and robust dielectric properties, has emerged as a promising filler for polymer-based dielectric composites. Nevertheless, its practical implementation in polymer matrices has been hindered by poor dispersibility and inadequate interfacial adhesion. This study addresses these challenges by developing hyperbranched polymer (HBP)-functionalised BN (HBP-BN) and fabricating HBP-BN/cellulose nanofiber (CNF) composite films via vacuum-assisted self-assembly. The HBP-BN/CNF films demonstrate a low dielectric loss ($\tan\delta < 0.016$) and significantly improved volume resistivity ($10^9 \Omega \cdot m$) and high thermal stability (55.36% residual mass at 800°C). Notably, the surface flashover voltage is elevated from 13.82 to 15.61 kV, demonstrating superior insulation capability. All these results highlight the promising potential of HBP-BN/CNF composite films for use as advanced dielectric materials, providing a novel approach to develop eco-friendly high-performance materials for high-voltage insulation systems in next-generation electronics and electrical devices.

1 | Introduction

The rapid advancement of high-voltage transmission networks, electrified transportation systems and high-power electronic devices has subjected dielectric materials to superimposed extreme operational stresses during service. These challenges encompass escalating thermal loads, intensified electric field strengths and complex three-dimensional field distributions. Such harsh service conditions demand enhanced performance in material systems, particularly regarding insulation reliability, thermal stability and environmental compatibility [1–3]. Conventional polymer dielectrics (e.g., polyethylene and epoxy resins) exhibit inherent limitations: their low intrinsic thermal

conductivity ($\kappa < 0.3 \text{ W/(m}\cdot\text{K)}$) aggravates Joule heating under high electric fields ($> 50 \text{ kV/mm}$), whereas weak interfacial interactions with inorganic fillers often result in premature dielectric breakdown [4, 5]. Addressing these challenges requires innovative material designs that synergistically integrate excellent dielectric performance, suppressed space charge accumulation and enhanced insulation performance [6, 7]. Composite films emerge as a groundbreaking solution through the strategic convergence of advanced nanomaterials and innovative structural engineering techniques. By harnessing the synergistic interplay between nanoscale material architectures and macroscopic design principles, these films transcend the inherent performance constraints of conventional dielectric

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insulators, establishing themselves as one of the most promising candidates for next-generation electrical insulation systems.

Boron nitride (BN), a graphene-isostructural III–V compound, has emerged as a transformative filler for dielectric composites owing to its unique combination of an ultra-wide bandgap (> 6 eV) and anisotropic thermal conductivity ($\kappa_{\parallel} = 300\text{--}400$ W/(m·K), $\kappa_{\perp} = 30\text{--}40$ W/(m·K)) [8, 9]. These properties theoretically enable BN-based composites to achieve exceptional dielectric strength (> 40 kV/mm) while efficiently dissipating thermal energy—a critical requirement for high-voltage insulation systems [10]. However, conventional hexagonal BN (h-BN) suffers from severe platelet aggregation due to strong interlayer van der Waals forces, leading to poor dispersion in polymer matrices and interfacial debonding under combined electrical–thermal stresses [11, 12].

Surface functionalisation via covalent grafting remains the predominant approach to improving BN dispersibility and interfacial interactions in polymer matrices. Conventional strategies, such as silane coupling agents [13, 14] or plasma treatments [15, 16], introduce functional groups (–OH, –NH₂) to enhance polymer–filler compatibility. For example, Wu et al. functionalised BN with amino groups by leveraging boron atom electron vacancies and nitrogen atom excess electrons in urea molecules [17]. Although these methods partially mitigate aggregation, they exhibit low grafting density (< 5 groups/nm²) and limited spatial configuration control [18, 19]. The resultant linear or planar molecular architectures fail to establish sufficient mechanical interlocking with polymer chains, particularly under simultaneous electrical–thermal stresses [20].

In contrast, hyperbranched polymer (HBP) grafting offers a three-dimensional topological structure with high-density terminal groups (> 20 groups/nm²), enabling concurrent enhancement of dispersibility and interfacial stress transfer [21]. The fractal geometry of HBP induces molecular-scale entanglement with matrix chains, whereas its abundant functional termini provide covalent anchoring points—a dual mechanism unattainable with conventional modifiers [22, 23]. This architecture not only strengthens interfacial adhesion but also preserves the intrinsic thermal and electrical properties of BN, overcoming the trade-offs inherent in traditional surface modifications.

Cellulose nanofibers (CNF), derived from renewable biomass, serve as an eco-friendly alternative to synthetic polymers, exhibiting remarkable mechanical strength (Young's modulus > 10 GPa) and intrinsic dielectric stability (breakdown strength > 40 kV/mm) [24–26]. However, the hydrophilicity and low thermal conductivity ($\kappa \approx$ W/(m·K)) of CNF restrict its application in high-voltage systems requiring moisture resistance and thermal management [27]. Integrating BN into CNF matrices presents a promising solution by combining high thermal conductivity ($\kappa \approx 300\text{--}400$ W/(m·K)) and wide bandgap (> 6 eV) of BN with mechanical robustness and dielectric stability of CNF. This synergy enhances dielectric and insulation performance through uniform charge distribution and increased surface flashover voltage [28].

In this study, we fabricate a high-performance composite film via vacuum-assisted filtration self-assembly of one-dimensional

(1D) CNF and two-dimensional (2D) BN. HBP mediated surface functionalisation optimises BN dispersion within the CNF matrix. HBP is selected for its highly branched architecture and abundant terminal functional groups, which not only modify BN surfaces but also establish multiple hydrogen bonds with CNF, significantly strengthening interfacial interactions. Systematic investigation reveals that the hyperbranched structure acts as a molecular bridge, forging robust CNF–BN interfaces, whereas the 1D/2D configuration facilitates efficient charge dissipation and thermal transport. These findings demonstrate that hyperbranched surface functionalisation provides a simple and effective strategy for integrating superior electrical and thermal properties into composites, advancing the development of tailored dielectric materials for high-voltage applications.

2 | Experiments

2.1 | Materials

Cellulose nanofibril was provided by the Institute of Zhejiang University-Quzhou, China. Hexagonal boron nitride (h-BN, 1–2 μ m) was provided by Shanghai Aladdin Bio-chemical Technology Co. Ltd., China. Accompanying chemicals such as 3,5-diaminobenzoic acid (DABA), pyridine, 3-aminopropyltriethoxysilane (KH-550) and triphenyl phosphoric acid ester (TPP), along with solvents and reagents including N-methyl pyrrolidone (NMP), lithium chloride (LiCl), acetic acid and ethyl alcohol were procured from Macklin Biochemical Co. Ltd, China.

2.2 | Preparation of BN–NH₂ and HBP–BN

Initially, 1 g of BN and 20 g of KH-550 were ultrasonically dispersed in 300 g of anhydrous ethanol for 60 min under ice-water bath conditions (0°C–5°C) to obtain hydroxylated BN–OH. The homogeneous suspension was transferred into a three-neck flask equipped with a condenser and subjected to reflux condensation at 80°C. Subsequently, a co-solvent mixture comprising 40 g deionised water and 100 g anhydrous ethanol was introduced dropwise into the reaction system. The silanisation reaction between BN–OH and KH-550 proceeded under continuous mechanical stirring at 80°C for 12 h (Figure 1). Post-reaction, the product was isolated via hot centrifugation (8000 rpm, 60°C) and sequentially washed eight times with deionised water to remove unreacted silane agents. The purified BN–NH₂ was finally lyophilised in a vacuum freeze dryer (–50°C, 10 Pa) for 48 h.

The HBP–BN was synthesised through a step-growth polymerisation approach adapted from established methodologies [29]. The preparation method of HBP–BN has been previously reported in our earlier work [21]. A schematic illustration of the synthesis process is provided in Figure 1. Briefly, 0.6 g of BN–NH₂ and 0.6 g of DABA were dispersed in 50 mL of NMP under vigorous stirring until complete dissolution. The reaction system was then supplemented with 2 mL of pyridine (catalyst) and 6 mL of TPP (stabilizer), followed by the addition of 0.2 g LiCl to modulate ionic strength. Polymerisation was conducted under nitrogen atmosphere at 100°C for 12 h with constant magnetic stirring (500 rpm). Upon reaction completion,

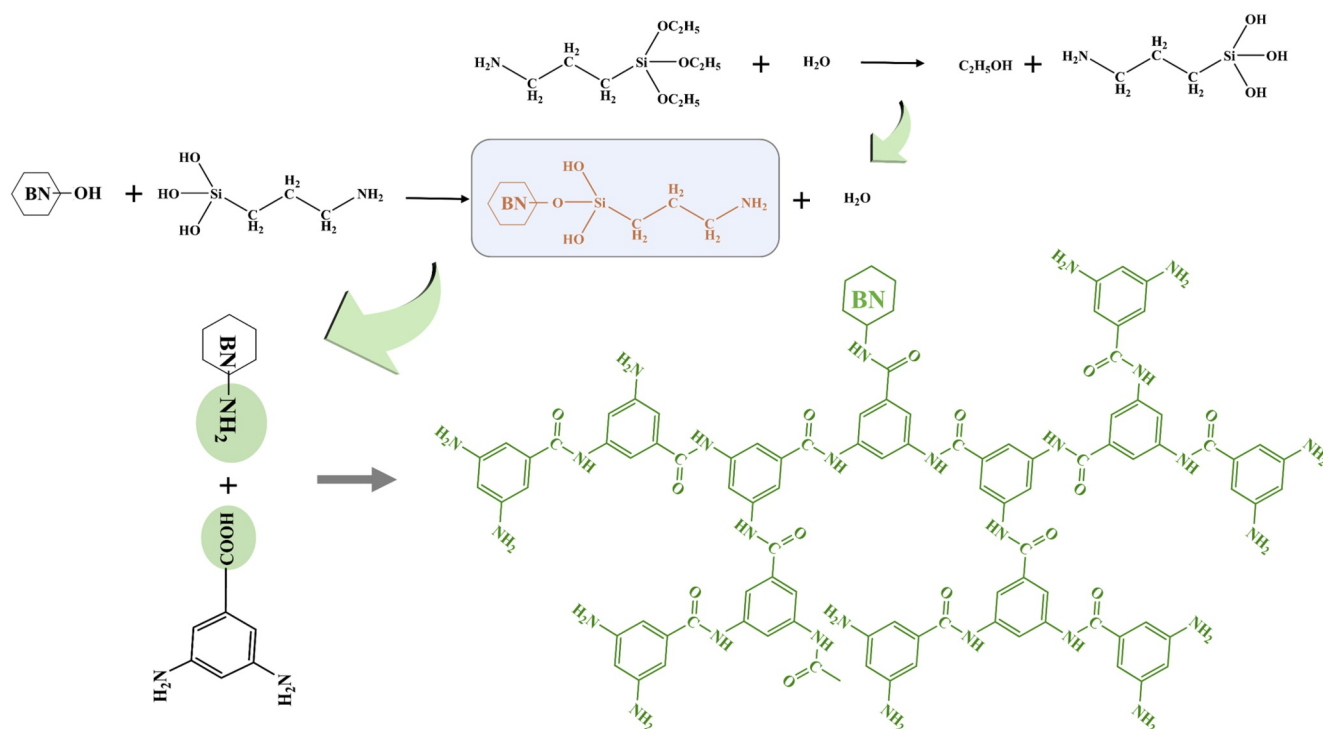


FIGURE 1 | Illustration of the synthesis route of the HBP-BN.

the crude product underwent hot centrifugation (10,000 rpm, 80°C) to separate grafted HBP-BN from residual monomers. Sequential washing with NMP (3×20 mL) ensured removal of unreacted DABA, and the final product was dried under vacuum freeze-drying conditions (-50°C , 10 Pa) for 48 h.

2.3 | Preparation of HBP-BN/CNF Composite Films

0.05 g of HBP-BN was dissolved in 30 mL of deionised water and ultrasonically dispersed for 20 min to ensure uniform distribution. Subsequently, 0.05 g of CNF suspension (1.69 wt%) was added to the dispersion, and the mixture was vigorously stirred to achieve a homogeneous HBP-BN/CNF suspension. The detailed fabrication process is schematically depicted in Figure 2. The resulting suspension was transferred to a sand core funnel and subjected to vacuum filtration. The filtered membrane was then dried in an oven to obtain the HBP-BN/CNF composite film. Following this procedure, composite films with HBP-BN contents of 0, 10, 20, 30, 40 and 50 wt% were fabricated, with the total mass of each film precisely controlled at 0.01 g.

2.4 | Characterisations

The HBP-BN was prepared following the same procedure as reported in our previous study [21], where detailed synthesis and characterisation protocols are provided. The chemical functionalisation of BN and HBP-BN was investigated by Fourier transform infrared spectroscopy (FTIR, Varian Excalibur 3100) with a spectral resolution of 4 cm^{-1} . Structural evolution was analysed via Raman spectroscopy (Renishaw inVia) across the $1000\text{--}2000\text{ cm}^{-1}$ range. X-ray photoelectron spectroscopy (XPS,

Thermo Scientific K-Alpha) employing Al K α radiation (1486.6 eV) provided elemental composition, chemical states and bonding configurations. Crystallographic features were examined by X-ray diffraction (XRD, Bruker D8 Advance) with Cu K α radiation ($\lambda = 1.5406\text{ \AA}$) at diffraction angle 2θ of $5^{\circ}\text{--}80^{\circ}$ (scan rate: $5^{\circ}\text{ min}^{-1}$). Thermogravimetric analysis (TGA, Netzsch STA 449 F3) under a nitrogen atmosphere ($30^{\circ}\text{C}\text{--}800^{\circ}\text{C}$, heating rate: 10°C/min) quantified the HBP grafting efficiency by mass loss differentials. Surface morphology was visualised using field-emission scanning electron microscopy (FE-SEM, Hitachi S-4800) at 5 kV accelerating voltage.

The microstructural characteristics of HBP-BN and CNF were systematically investigated using transmission electron microscopy (TEM, JEOL JEM-F200, Japan) to resolve nanoscale surface morphologies. Cross-sectional analysis of the HBP-BN/CNF composite films was performed via field-emission scanning electron microscopy (SEM, ZEISS Sigma 360, Germany). Thermogravimetric analysis (TGA, Netzsch STA449F5, Germany) was conducted under nitrogen atmosphere (20 mL/min, $25^{\circ}\text{C}\text{--}800^{\circ}\text{C}$, 10°C/min) to assess thermal stability. Derivative thermogravimetry (DTG) curves were analysed to determine decomposition kinetics and residual mass profiles under programmed heating conditions. Dielectric properties, including frequency-dependent permittivity (ϵ') and loss tangent ($\tan\delta$), were measured using a precision LCR metre (Agilent E4980A) across $10\text{--}10^7\text{ Hz}$ at 25°C . Volume resistivity was determined via high-resistance metre (Keithley 6517B).

The flashover voltage was measured using a DW-P603-3ADC2 positive-polarity high-voltage DC power supply. The experiments were conducted under controlled environmental conditions, maintaining a constant temperature of $(27 \pm 3)^{\circ}\text{C}$ and relative humidity of $(40 \pm 5)\%$. Brass finger electrodes with a fixed spacing of 12 mm were employed for the measurements.

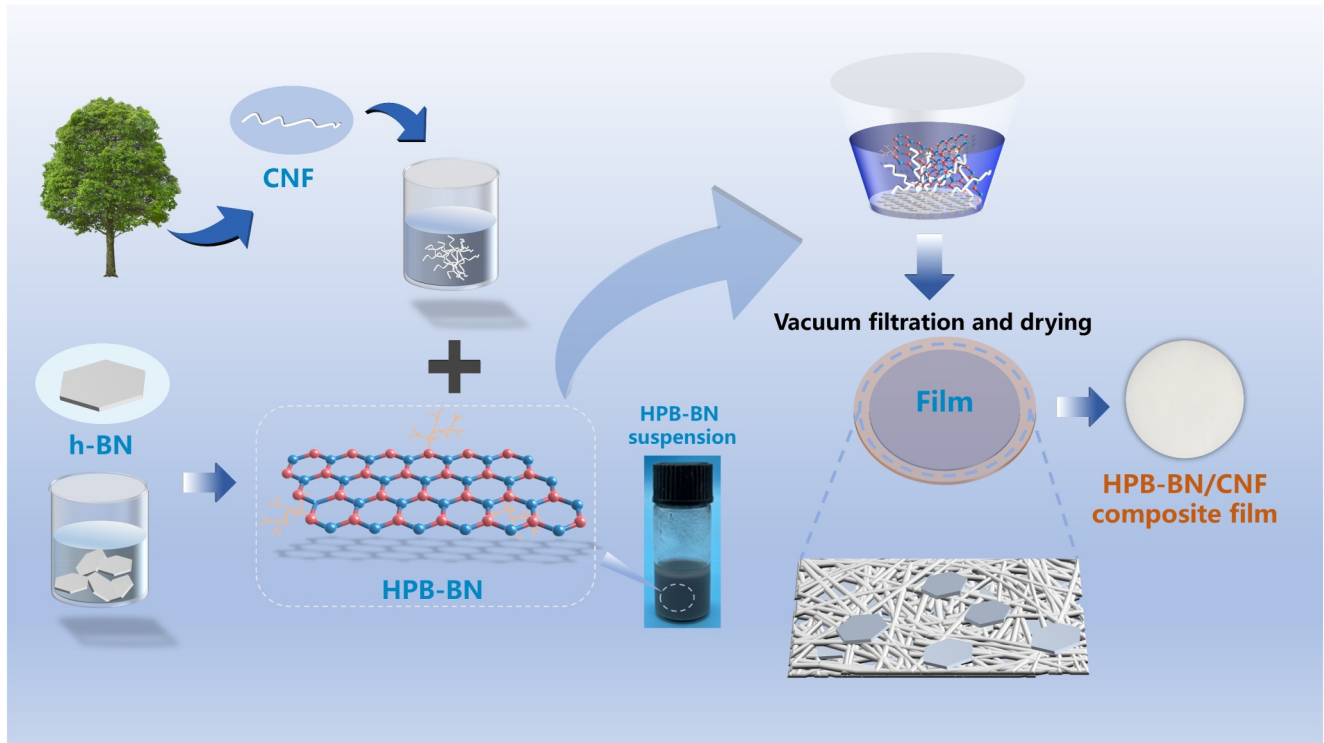


FIGURE 2 | Schematic illustration of the fabrication process for HBP-BN/CNF composite films.

For each sample, the test was repeated 10 times, and the average value was recorded as the flashover voltage. To ensure the elimination of residual surface charges, a 5-min interval was maintained between consecutive discharges.

The surface potential decay (SPD) characteristics of the samples were evaluated using the SPD method. A stainless steel needle connected to a high-voltage DC power supply served as the needle electrode, with the electrode voltage set to +5 kV. After voltage adjustment, the samples were subjected to corona charging for 10 min. Following the power shutdown, the potential signals were monitored and recorded in real time using a Kelvin probe. The obtained data were used to plot the SPD curves, with a total recording duration of 40 min. A carrier trap is a phenomenological concept of the charge transport characteristics of a dielectric medium. The method of SPD indicates that the characteristic parameters of the trap distribution, including the trap depth E_T and the trap density $I(t)$, can be calculated from the curves of the surface potential decaying [30].

$$E_T = \kappa_B T \ln(\nu t), \quad (1)$$

$$I(t) = \left(\frac{\epsilon_0 \epsilon_r}{e T_h} \right) \cdot \frac{tdU}{dt}, \quad (2)$$

where κ_B is the Boltzmann constant, T is the temperature, t is the time, T_h is the sample thickness and ν is the charge escape frequency, which is $4.17 \times 10^{13} \text{ s}^{-1}$ for a DC field. e is the elementary charge. dU/dt is the decay rate of the surface potential.

These multi-modal characterisation approaches provide comprehensive insights into the structure–property relationships critical for optimising composite film performance in next-generation electronics and electrical devices.

3 | Results and Discussion

3.1 | Synthesis and Characterisation of HBP-BN

The ultrasonic exfoliation process effectively reduced the lateral dimensions of BN to submicron scales while producing ultrathin lamellar structures, as evidenced by the increased transparency in SEM images (Figure 3a and Supporting Information S1: Figure S1) [31]. High-resolution TEM characterisation revealed well-defined lattice fringes with 0.224 nm interplanar spacing, corresponding to the (100) crystallographic plane of h-BN (Figure 3b), confirming maintained structural integrity during the exfoliation process. This preserved crystalline architecture ensures efficient phonon transport pathways, a critical factor for thermal management in dielectric composites. Simultaneously, CNFs displayed characteristic high-aspect-ratio morphology with 5–10 nm diameters and micrometre-scale lengths, as observed in TEM analysis (Figure 3c,d). The extended nanofibril geometry facilitates three-dimensional percolative network formation through interlocking fibril junctions, creating mechanically robust frameworks capable of stress redistribution. This hierarchical architecture synergistically enhances electrical insulation properties by establishing tortuous charge carrier pathways while minimising interfacial defect formation.

The covalent functionalisation of BN through HBP grafting was unequivocally confirmed by complementary spectroscopic analyses. FTIR spectra revealed critical chemical modifications, where HBP-BN exhibited four emergent absorption bands compared to pristine BN (Figure 4a). A broad peak at 3220 cm^{-1} corresponds to N-H bending vibrations, whereas the distinct signature at 2925 cm^{-1} arises from C-H stretching in methylene ($-\text{CH}_2-$) groups. The appearance of peaks at 1631 and 1139 cm^{-1}

further corroborates the formation of amide (-CO-NH-) linkages and siloxane (-Si-O-) bridges, respectively, confirming successful HBP anchoring via silane coupling agents. XPS provided atomic-level validation of this surface engineering process. Survey spectra (Figure 4b) demonstrated the emergence of Si2s (154 eV) and Si2p (103 eV) signals in both BN-NH₂ and HBP-

BN, verifying effective silanisation through KH-550 treatment. High-resolution C1s deconvolution (Figure 4c) resolved four chemical states: carboxyl (O-C=O at 288.5 eV), amine-bound carbon (C-N at 285.5 eV), aromatic structures (C=C at 284.8 eV) and aliphatic chains (C-C at 284.2 eV), collectively mapping the molecular architecture of HBP. Concurrently, the Si2p spectrum (Figure 4d) differentiated two bonding environments -Si-O-C (102.79 eV) and Si-C (102.14 eV) elucidating the dual interfacial anchoring mechanisms. These spectral signatures align precisely with the proposed reaction pathway in Figure 1, where silanol groups from KH-550 undergo condensation with BN surface hydroxyls, followed by nucleophilic attack from HBP terminal amines. The synergistic FTIR/XPS evidence establishes a chemically robust BN-HBP interface, fundamentally explaining the enhanced dielectric stability of composite through suppressed interfacial polarisation and optimised phonon/charge transport pathways.

TGA systematically validated the covalent functionalisation of BN with HBP, revealing critical insights into the thermal stability and grafting efficiency of the hybrid system (Supporting Information S1: Figure S2). The pristine BN exhibited minimal mass loss (4%) across the 30°C–800°C range, attributable to surface-bound hydroxyl group dehydration and adsorbed water elimination. In contrast, HBP-BN displayed characteristic three-stage decomposition behaviour: initial solvent evaporation (below 200°C), followed by polymer backbone degradation (between 200°C and 625°C) and final carbon residue oxidation

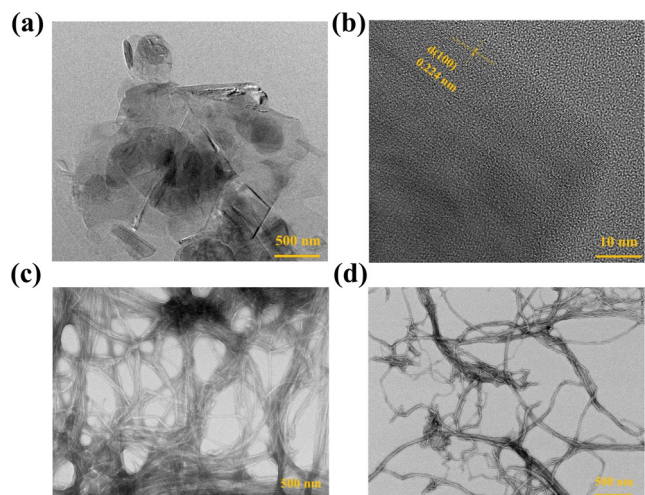


FIGURE 3 | TEM images of samples. (a) TEM image of BN, (b) HRTEM image of BN, (c) TEM images of CNFs (200 nm) and (d) TEM images of CNFs (500 nm).

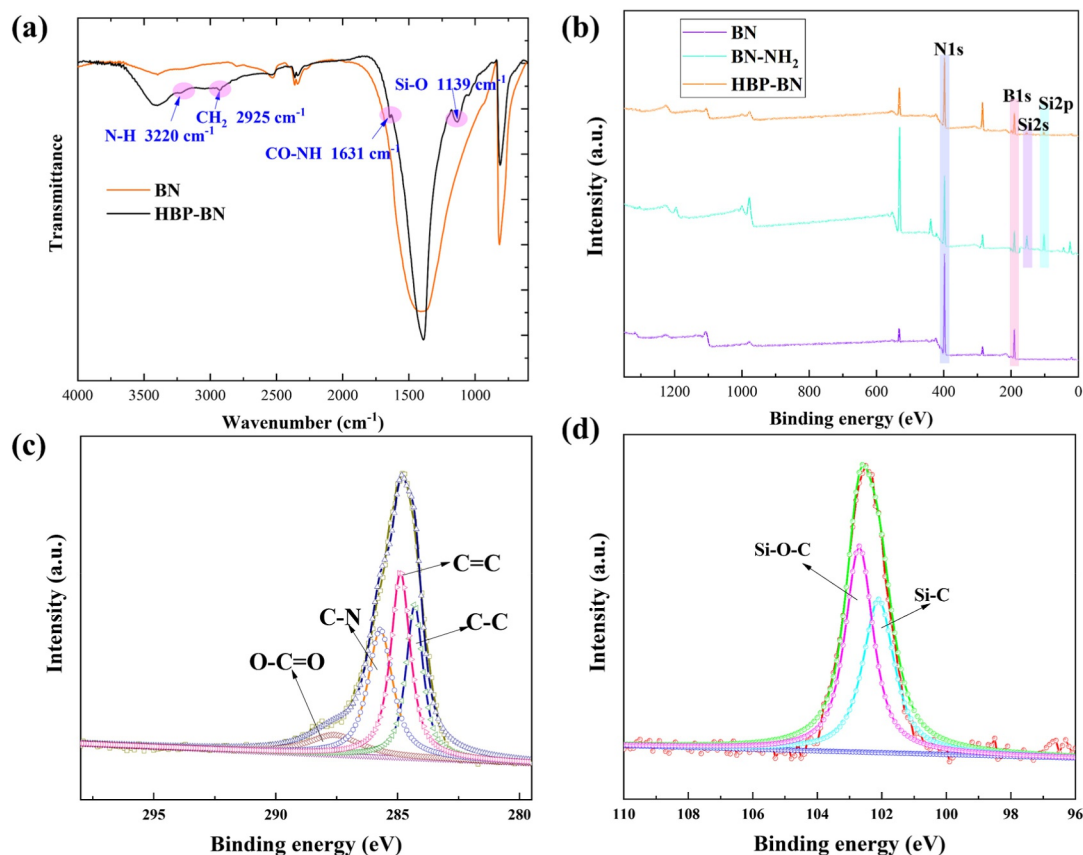


FIGURE 4 | The characterisation results before and after BN modification. (a) FTIR result of BN and HBP-BN, (b) XPS images of BN and HBP-BN, (c) C1s spectra of HBP-BN and (d) Si2p spectra of HBP-BN.

(< 0.5% above 625°C). This staged decomposition profile aligns with the thermal degradation mechanisms of hyperbranched architectures, where β -scission of ester linkages precedes amide bond cleavage in branched polymer chains [32].

The grafting ratio was determined to be 17% following buoyancy correction via parallel blank experiments, calculated from the differential mass loss between HBP-BN (21%) and pristine BN (4%). This methodology employs standardised thermogravimetric protocols for hybrid material systems, effectively isolating the organic phase contribution while considering the intrinsic thermal stability of inorganic matrices [14]. Post-thermal treatment XRD patterns confirmed retention of the hexagonal crystal structure of BN, with no detectable signatures of graphitic carbon (Supporting Information S1: Figure S3), thereby verifying complete HBP elimination without structural compromise of the ceramic framework. X-ray diffraction analysis (Supporting Information S1: Figure S3) systematically elucidates the crystallographic evolution of BN and HBP-BN. The pristine BN exhibits characteristic diffraction peaks at 26.9° (002), 41.6° (100), 55.1° (004) and 76.2° (110), which are precisely indexed to the h-BN structure. Notably, HBP-BN maintains identical peak positions with preserved peak intensity ratios, confirming the structural integrity of the hexagonal framework during the grafting process. The sharp diffraction profiles demonstrate minimal lattice distortion despite polymer functionalisation. In addition, a distinct emergent peak at 31.3° suggests interfacial bonding formation, potentially corresponding to covalent B-O-C linkages between BN and HBP. The absence of amorphous halos in the XRD pattern further confirms the ordered nature of the BN-HBP interface.

3.2 | Morphological Characterisation

Having demonstrated the successful preparation of efficient functionalisation of HBP-BN, we constructed an HBP-BN/CNF composite film system with controllable microstructures via a vacuum filtration-assisted self-assembly strategy (Figure 2). This methodology achieves ordered assembly of HBP-BN nanosheets within the CNF matrix through precise regulation of shear fields and dynamic molecular interactions. Cross-sectional SEM analysis (Figure 5a–f) systematically reveals the structural evolution governed by filler content: The pure CNF film (Figure 5a) displays a characteristic hierarchical fibre network with anisotropic alignment and homogeneous porosity formed through hydrogen bonding. Upon incorporating HBP-BN (Figure 5b–f), the functionalised nanosheets align axially along CNF fibres, forming a brick-and-mortar architecture. Notably, even at 50 wt% loading (Figure 5f), HBP-BN maintains in-plane orientation without aggregation, preserving uniform interlayer spacing. This exceptional dispersion originates from a triple regulatory mechanism of hyperbranched polymers: (1) High-strength hydrogen bonding between terminal polar groups and CNF hydroxyls; (2) steric hindrance from three-dimensional topological configurations and (3) orientation assembly guided by π - π interactions. Their synergistic effects overcome the conventional dispersion-performance trade-off in nanocomposites [21, 22]. The HBP critically enhances BN dispersion in CNFs through its three-

dimensional topological structure and high grafting density, which synergistically suppress van der Waals-driven BN aggregation at the molecular level. The branched architecture generates spatial steric hindrance that physically separates BN nanosheets, directly evidenced by uniform dispersion without aggregation at 50 wt% loading (Figure 5f)—a condition where non-functionalised BN composites exhibit severe agglomeration. Concurrently, HBP terminal polar groups form hydrogen bonds with CNF hydroxyls, improving interfacial compatibility and enabling BN integration into the CNF network. This is further reinforced by π -interactions between BN planes and aromatic domains in lignin-containing CNF, which guide axial alignment along CNF fibres (Figure 5b–e). Collectively, these mechanisms establish an entropy-driven dispersion regime during vacuum self-assembly, where HBP's covalent tethering to BN surfaces and directional interactions with CNF optimise interfacial energetics to achieve structural homogeneity. The synergy between hydrogen bonding and π - π stacking is further corroborated by the composite's dielectric stability ($\tan\delta < 0.016$), where charge leakage would arise from interfacial defects if these interactions were ineffective.

High-resolution SEM characterisation (Figure 5d1–f1) confirms molecular-level interpenetrating structures at HBP-BN/CNF interfaces, where continuous transition zones form between nanosheet edges and fibre surfaces. This unique interfacial architecture endows the composite films with remarkable macroscopic flexibility (Figure 6), enabling repeated bending and folding without structural failure. By synergistically integrating interface engineering and structural control, this work demonstrates a multiscale ordered assembly paradigm for inorganic–organic nanocomposites, establishing new design principles for advanced flexible functional materials.

3.3 | Dielectric Properties

The frequency-dependent dielectric loss tangent ($\tan\delta$) curves demonstrate a systematic correlation between HBP-BN content and dielectric dissipation behaviour in CNF composites (Figure 5g). Pure CNF exhibits the lowest $\tan\delta$ values across the entire frequency spectrum, consistent with its homogeneous hydrogen-bonded network that effectively suppresses interfacial polarisation. The characteristic $\tan\delta$ profile of CNF reveals interfacial polarisation induced by hydroxyl groups (-OH) and hydrogen-bonded networks at low frequencies (< 10^4 Hz), leading to increased dielectric loss with decreasing frequency, whereas high-frequency regions (> 10^4 Hz) show gradual loss stabilisation due to polarisation relaxation lagging behind electric field variations. All HBP-BN–CNF composite films maintain $\tan\delta$ trends similar to pure CNF, confirming the persistent dominance of the CNF matrix.

Elevated $\tan\delta$ values at low frequencies (< 10^4 Hz) with increasing HBP-BN content arise from multiscale synergistic mechanisms. The interpenetrating HBP-BN/CNF architectures (Figure 5d–f) significantly enhance heterogeneous interfacial density, intensifying Maxwell–Wagner–Sillars (MWS) interfacial polarisation effects. This promotes interfacial bound charge

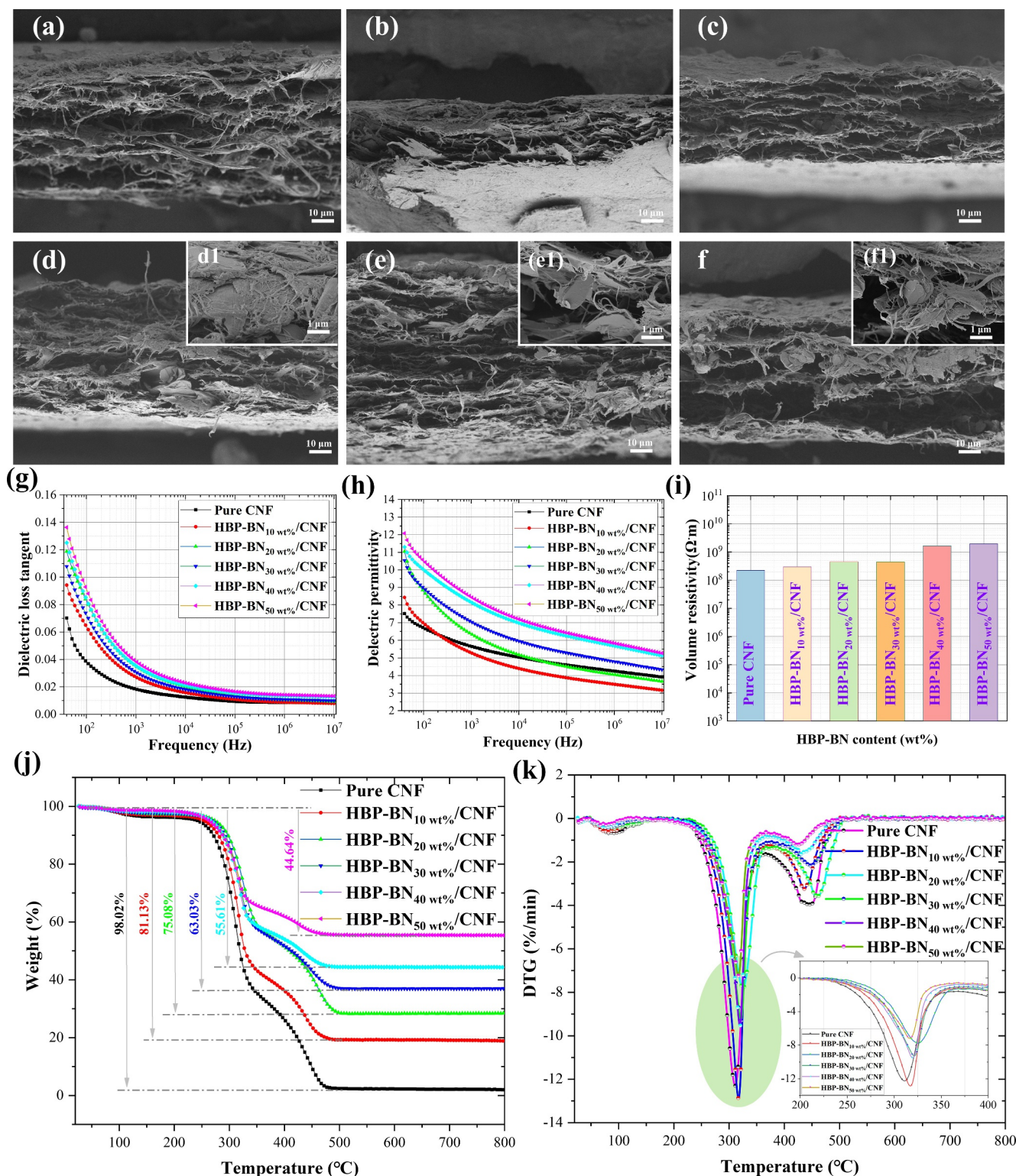


FIGURE 5 | Performance characterisation results of HBP-BN/CNF composite films. (a) Pure CNF, (b) HBP-BN₁₀ wt%/CNF, (c) HBP-BN₂₀ wt%/CNF, (d) HBP-BN₃₀ wt%/CNF, (e) HBP-BN₄₀ wt%/CNF and (f) HBP-BN₅₀ wt%/CNF. High-resolution SEM images of (d1) HBP-BN₃₀ wt%/CNF, (e1) HBP-BN₄₀ wt%/CNF and (f1) HBP-BN₄₀ wt%/CNF. (g) Dielectric loss tangent, (h) dielectric permittivity, (i) volume resistivity, (j) TGA curve and (k) DTG curve of pure CNF film and HBP-BN/CNF composite films.

accumulation and relaxation as the primary low-frequency energy dissipation pathway. Concurrently, unreacted polar groups (e.g., -OH and -NH₂) on HBP establish continuous ion migration channels at high filler loadings, activating low-frequency ionic

conduction losses. Furthermore, BN nanosheets act as physical barriers to prolong charge carrier migration paths, exacerbating energy dissipation through shallow trap-assisted hopping mechanisms [33].

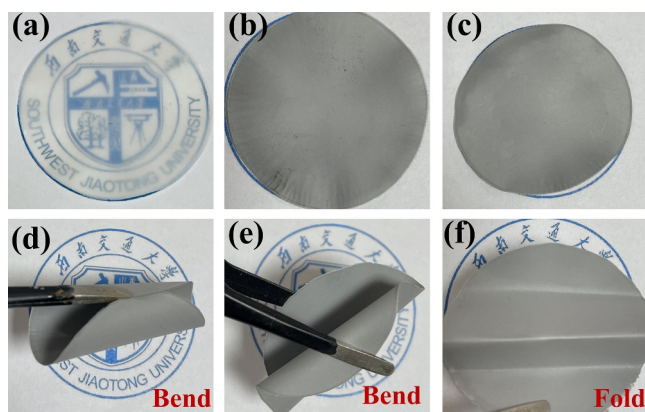


FIGURE 6 | The photographs of composite films and mechanical flexibility of the HBP-BN/CNF composite films. (a) Pure CNF, (b) HBP-BN₁₀ wt%/CNF, (c) HBP-BN₂₀ wt%/CNF, (d) HBP-BN₃₀ wt%/CNF, (e) HBP-BN₄₀ wt%/CNF and (f) HBP-BN₅₀ wt%/CNF.

In high-frequency regimes ($> 10^4$ Hz), $\tan\delta$ values remain below 0.016 despite increasing HBP-BN content, primarily attributed to intrinsic polarisation dynamics and material characteristics. Electron displacement polarisation with sub-nanosecond response times dominates under high-frequency electric fields, whereas dipole polarisation and interfacial effects induced by fillers are suppressed because of relaxation time mismatch. The wide bandgap of BN (~ 5.9 eV) and its high crystallinity (XRD-confirmed (002) interplanar spacing of 0.334 nm) fundamentally restrict intrinsic carrier concentration, effectively blocking Ohmic conduction pathways. Additionally, steric hindrance from HBP substantially reduces defect state density at HBP-BN/CNF interfaces, diminishing high-frequency localised state-assisted quantum tunnelling probabilities. This interfacial optimisation aligns with SEM-observed continuous transitional structures (Figure 5d1–f1), corroborating the microstructure-property relationship.

As shown in Figure 5h, the dielectric permittivity–frequency curves of HBP-BN/CNF composite films exhibit a distinct concentration-dependent evolution [34, 35]. The similarity in curve profiles and parallel decay trends across different filler loadings indicate a unified polarisation mechanism, where variations in filler content primarily modulate dielectric response intensity through interfacial density rather than altering the fundamental polarisation dynamics. Pure CNF films display stable dielectric behaviour across the entire frequency range, with minor frequency dependence attributed to dipole relaxation of -OH in CNF molecular chains and weak interfacial polarisation within the hydrogen-bonded network. Upon incorporation of HBP-BN, the dielectric behaviour demonstrates a nonlinear concentration effect: the HBP-BN₁₀ wt%/CNF composite film exhibits lower dielectric permittivity than pure CNF at all frequencies, likely due to a dual suppression mechanism [34, 36]. The wide bandgap (~ 5.9 eV) of HBP-BN impedes ion migration polarisation inherent in the CNF matrix, whereas isolated HBP-BN nanosheets introduce interfacial defects that distort local electric fields, further diminishing polarisation efficiency.

When the HBP-BN content increases to 20 wt%, the dielectric permittivity in the low-frequency region ($< 10^4$ Hz) surpasses

that of pure CNF, whereas the high-frequency response ($> 10^4$ Hz) remains slightly lower. This frequency-dependent crossover suggests a competitive polarisation mechanism: MWS interfacial polarisation induced by heterogeneous interfaces dominates at low frequencies, while rapid electronic displacement polarisation in the CNF matrix at high frequencies is mildly suppressed due to restricted molecular chain mobility caused by filler incorporation [37].

At HBP-BN loadings of 30 wt% and above, the dielectric permittivity of HBP-BN/CNF composites exhibits systematic enhancement with strictly parallel curve profiles. This transition stems from the complete formation of an interfacial polarisation network. Strong coupling between -OH on HBP-BN surfaces and the CNF hydrogen-bonded network promotes uniform dispersion of HBP-BN nanosheets (as evidenced by agglomeration-free SEM images in Figure 5b–f), creating a high density of heterogeneous interfaces. Additionally, nanoscale micro-capacitors formed between adjacent HBP-BN nanosheets amplify polarisation through localised electric field superposition, while the high crystallinity of BN layers acts as physical barriers, prolonging charge carrier migration paths. This forces charge transport to rely on shallow trap-assisted hopping mechanisms, strictly confining polarisation enhancement to interfacial bound charge accumulation [35]. Notably, even at high HBP-BN loadings (50 wt%), the dielectric permittivity increases steadily without percolation threshold characteristics, confirming that HBP modification inhibits conductive network formation by blocking direct filler contact.

Furthermore, the parallel decay of dielectric permittivity with increasing frequency across all samples indicates that polarisation response kinetics are governed by the intrinsic relaxation properties of both the matrix and interfacial polarisation [36, 38]. In the high-frequency region ($> 10^4$ Hz), the behaviour is particularly illustrative: despite the significantly higher absolute dielectric permittivity of the HBP-BN₅₀ wt%/CNF composite film compared to pure CNF film, its decay slope aligns closely with low-loading samples. This confirms that rapid electronic displacement polarisation remains the dominant high-frequency mechanism, whereas HBP-BN contributes supplementary polarisation through micro-capacitor coupling (local charge redistribution between adjacent nanosheets). The stability of these polarisation dynamics ensures predictable performance, enabling precise dielectric constant tuning via filler content—a critical foundation for gradient design in flexible devices.

Comparisons with unmodified BN/polymer systems further highlight the pivotal role of interface engineering. Conventional BN nanosheet composites often exhibit abrupt dielectric constant surges at 20 wt%–30 wt% because of percolation effects [39]. In contrast, the gradual enhancement trend and low loss characteristics observed here confirm that HBP modification suppresses electron tunnelling between fillers through chemical and steric effects, ensuring polarisation enhancement is solely governed by interfacial mechanisms. This performance decoupling strategy, achieved through tailored surface chemistry and microstructure design, provides a novel pathway for developing advanced composites with high permittivity, low loss and mechanical flexibility.

The volume resistivity of the HBP-BN/CNF composite films exhibited a marked increase with higher HBP-BN loadings, further corroborating the dominance of interfacial polarisation mechanisms (Figure 5i). Pure CNF films displayed moderate resistivity due to limited charge carrier mobility within their hydrogen-bonded network. The incorporation of HBP-BN significantly enhanced insulating properties through synergistic mechanisms: the wide bandgap (~ 5.9 eV) of HBP-BN, coupled with the insulating HBP surface modification, fundamentally suppressed the generation and migration of free charges. Concurrently, the uniformly dispersed BN nanosheets established a three-dimensional insulating barrier network within the matrix, forcing charge carriers to traverse highly tortuous pathways, resulting in an exponential decline in mobility with increasing filler content. The nanoscale micro-capacitor effect between adjacent BN nanosheets amplified interfacial polarisation through localised bound charge accumulation while preventing long-range charge transport, thereby synergistically enhancing dielectric constant and suppressing conduction loss. At 50 wt% HBP-BN loading, the volume resistivity increased by over one order of magnitude compared to pure CNF, aligning with the dielectric enhancement trend. This consistency demonstrates that the formation of interfacial polarisation networks not only optimises dielectric response but also synergistically inhibits charge leakage through physical barrier effects and chemical modification. The HBP-BN/CNF composite films achieve exceptional dielectric-insulation performance through hyperbranched polymer-mediated interfacial engineering, where the three-dimensional branched topology of HBP enables synergistic hydrogen bonding (between terminal polar groups $-\text{OH}/-\text{COOH}$ and CNF hydroxyls) and π - π stacking, directly evidenced by uniform dispersion at 50 wt% loading (Figure 5f) and axial nanosheet alignment along CNF fibres (Figure 5b-e). These interactions suppress aggregation and optimise stress transfer, yielding > 10 -fold volume resistivity enhancement, elevated flashover voltage (15.61 kV) and stable dielectric properties ($\epsilon = 5.8$, $\tan\delta < 0.016$ at 10 MHz) through dual-trap energy regulation. The concurrent improvement in both insulating performance and dielectric permittivity provides a novel design paradigm for advanced applications demanding high dielectric strength and signal stability, such as high-voltage insulation and high-frequency flexible electronics.

3.4 | Thermal Stability

The synergistic analysis of TGA (Figure 5j) and derivative thermogravimetric (DTG) curves (Figure 5k) comprehensively reveals the multistage thermal degradation behaviour of HBP-BN/CNF composite films and their correlation with filler concentration. As shown in Figure 5j, pure CNF films exhibited a mass loss of 98.02% at 800°C, whereas the incorporation of HBP-BN progressively reduced this mass loss to 81.13%, 75.08%, 63.03%, 55.61% and 44.64% for composites with 10 wt%–50 wt% HBP-BN. This gradient trend confirms the critical role of HBP-BN in enhancing thermal stability, a mechanism further elucidated by the distinct degradation stages observed in the DTG profiles. The DTG curves in Figure 5k reveal three characteristic degradation stages. A minor peak at 75°C–85°C is attributed to the desorption of adsorbed moisture or residual solvents, with

its negligible peak area indicating minimal volatile impurities and no significant dependence on filler concentration. The dominant peak at 311°C–327°C corresponds to the decomposition of CNF cellulose chains and HBP components. Notably, the peak area proportion systematically decreases with increasing HBP-BN content, demonstrating that uniformly dispersed BN nanosheets delay matrix degradation by prolonging heat diffusion pathways and suppressing volatile release. A secondary peak at 422°C–462°C, likely associated with the oxidation of residual carbonaceous products or complete decomposition of BN surface modifiers, shifts towards higher temperatures at elevated filler loadings, suggesting that stabilised interfacial phases between BN and the matrix inhibit high-temperature oxidative reactions. Despite the decomposition of HBP components within 200°C–400°C (second peak), the high thermal inertness of BN nanosheets ($> 1000^\circ\text{C}$ decomposition temperature) governs the final residual mass. The enhanced thermal stability of the HBP-BN/CNF composite films (55.36% residual mass at 800°C) primarily stems from synergistic interactions during thermal degradation. BN inherently resists oxidation up to 800°C, while the HBP undergoes controlled aromatisation to form a condensed carbonaceous char that encapsulates BN, creating a physical barrier against oxygen diffusion and volatile release. Simultaneously, multi-point hydrogen bonding between HBP terminal groups ($-\text{OH}/-\text{COOH}$) and CNF hydroxyls delays CNF decomposition by stabilising glycosidic bonds against pyrolysis. The BN nanosheets further shield the CNF matrix from direct thermal radiation, and their high thermal conductivity dissipates localised heat. This triple protection—char barrier formation, interfacial bonding reinforcement and nanosheet shielding—collectively suppresses oxidative degradation pathways without requiring protective surface layers on BN itself, as validated by the composite's residual mass exceeding theoretical predictions for physical mixtures. For instance, the 50 wt% HBP-BN/CNF composite shows a mass loss of 44.64% (vs. 98.02% for pure CNF), with residual mass (55.36%) primarily originating from thermally stable BN contributions. The evolution of DTG peak profiles further validates filler dispersion states: The narrowed and stabilised main peak (311°C–327°C) aligns with the agglomeration-free BN distribution observed in SEM images (Figure 5b-f), whereas filler aggregation typically induces peak splitting or abnormal temperature shifts. This retardation effect is further enhanced by the intrinsically high thermal conductivity of BN, which facilitates rapid heat dissipation to minimise localised thermal accumulation, evidenced by the narrowed full-width-at-half-maximum of DTG peaks. Although covalent bonding between HBP-BN and CNF is not explicitly observed at processing temperatures, interfacial interactions—including physical entanglement of hyperbranched polymer chains with cellulose fibrils and potential hydrogen bonding between BN edge groups and CNF hydroxyls—contribute to structural integrity during thermal stress. Meanwhile, the dominance of physical mechanisms is corroborated by the close alignment between experimental residual mass (55.36% at 50 wt% loading) and theoretical BN content (52.89%), confirming the chemically inert nature of BN governs thermal stability. To further evaluate the mechanical performance of HBP-BN/CNF composite films at elevated temperatures, Supporting Information S1: Figure S4 presents the dynamic mechanical analysis (DMA) results, with detailed analysis provided in the Supporting Information.

The interplay between thermal stability and dielectric performance highlights the dual functionality of HBP-BN. Whereas enabling precise dielectric permittivity tuning, BN nanosheets act as physical barriers to suppress both thermal degradation and charge leakage, complemented by HBP-mediated interfacial homogeneity that prevents localised heat accumulation. This synergy ensures minimal mass loss (< 5%) and broadband dielectric stability (1 kHz–1 MHz) within the operational temperature range of flexible electronics (< 150°C), positioning these composites as promising candidates for high-frequency applications like 5G circuit packaging. Future studies could focus on designing flame-retardant hyperbranched polymers or optimising BN surface functionalisation to tailor mid-to-high temperature (300°C–500°C) degradation pathways, thereby extending their utility in extreme environments.

3.5 | Carrier Trap Characteristics of HBP-BN/CNF Films

When the SPD curves are fitted with a double exponential function, the calculated trap curves always show two kinds of characteristic peaks [40]. Specifically, the trap energy levels were calculated using a rigorously standardised approach based on this double-exponential fitting, resolving charge detrapping kinetics into distinct fast-decay (shallow traps) and slow-decay (deep traps) phases. Trap depths were quantified via Arrhenius analysis of temperature-dependent detrapping time constants, consistently applied across all samples—shallow traps (0.761–0.802 eV) originate from hydroxyl defects and hydrogen bond cleavage in CNF, whereas deep traps (0.845–0.861 eV) arise from interfacial barriers induced by HBP-BN, as resolved in Figure 7a–h. Trap densities were derived from relative amplitude contributions normalised to unit film thickness and electrode area. As illustrated in Figure 7, the SPD analysis of HBP-BN/CNF composite films reveals a strong correlation between charge transport behaviour and interfacial trap energy levels. The double-exponential fitting of the decay curves identifies synergistic mechanisms between shallow traps (0.761–0.802 eV) and deep traps (0.845–0.861 eV), as shown in Figure 7a. For pure CNF (Figure 7c), the shallow trap energy (0.761 eV) and deep trap energy (0.845 eV) originate from hydroxyl defects and hydrogen bond cleavage in cellulose molecular chains. The shallow trap distribution facilitates rapid charge release, while the limited deep trap density restricts long-term bulk charge retention. With the incorporation of HBP-BN, the shallow trap energy exhibits a fluctuating upward trend with increasing HBP-BN content [41], whereas the deep trap energy shows only a slight enhancement compared to pure CNF, indicating that the incorporation of filler has limited regulatory effects on deep charge traps, as evidenced in Figure 7b.

At low HBP-BN loadings (10 wt%–20 wt%, Figure 7d,e), the increase in shallow trap energy may arise from intensified interfacial polarisation. The introduction of BN nanosheets elevates the density of heterogeneous interfaces within the CNF matrix, inducing localised electric field distortion and forming shallow traps with marginally higher energy levels (< 0.8 eV). Although these shallow traps moderately delay charge release, their low energy limits surface charge capture efficiency,

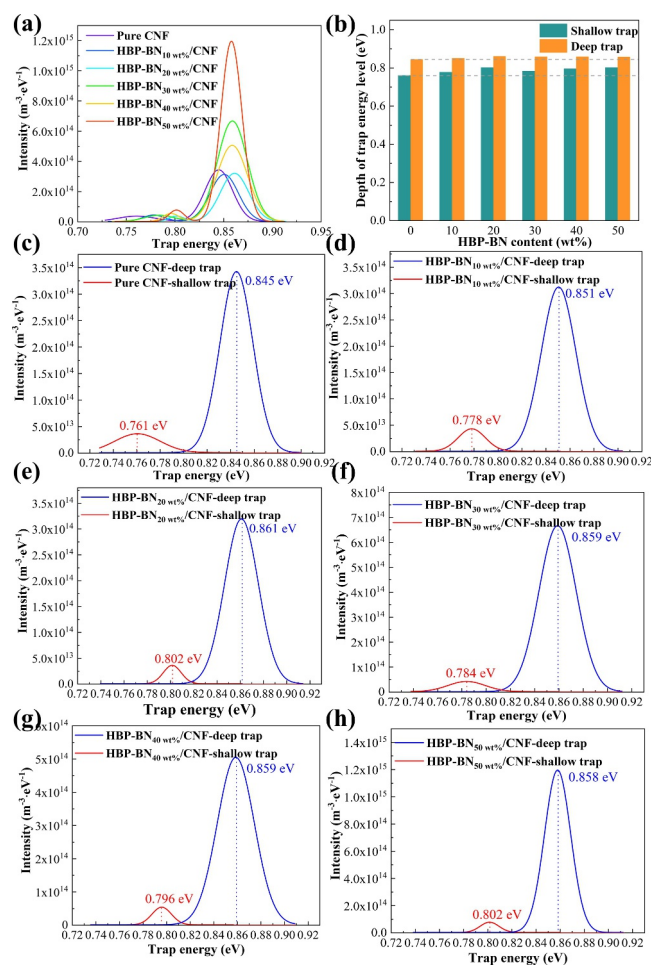


FIGURE 7 | Carrier trap distribution results. (a) The carrier trap distributions of HBP-BN/CNF composite films, (b) depth of trap energy level of HBP-BN/CNF composite films, (c) trap distributions of pure CNF, (d) trap distributions of HBP-BN₁₀ wt%/CNF, (e) trap distributions of HBP-BN₂₀ wt%/CNF, (f) trap distributions of HBP-BN₃₀ wt%/CNF, (g) trap distributions of HBP-BN₄₀ wt%/CNF and (h) trap distributions of HBP-BN₅₀ wt%/CNF.

allowing partial charges to dissipate rapidly. Concurrently, the slight rise in deep trap energy suggests that BN nanosheets prolong carrier migration paths via a physical barrier effect [42], forcing charges to be captured by deeper traps, thereby suppressing leakage currents. This aligns with the improved volume resistivity observed in Figure 5i. A notable anomaly occurs at 30 wt% HBP-BN (Figure 7f), where the shallow trap energy reaches a minimum (0.784 eV) while the deep trap energy peaks (0.861 eV). This phenomenon likely stems from a critical transition in filler dispersion: at 30 wt%, hyperbranched polymer chains form a dense interfacial layer through molecular extension, optimising BN-CNF compatibility and reducing shallow trap energy. The dynamic equilibrium between rapid charge dissipation via shallow traps and efficient charge immobilisation by deep traps synergistically enhances surface charge dissipation and bulk charge suppression [41], correlating with the peak flashover voltage (14.94 kV, Figure 8). When the HBP-BN content increases to 40 wt%–50 wt% (Figure 7g,h), the shallow trap energy rebounds to 0.796–0.802 eV, whereas the deep trap energy stabilises. This trend implies that as the filler content

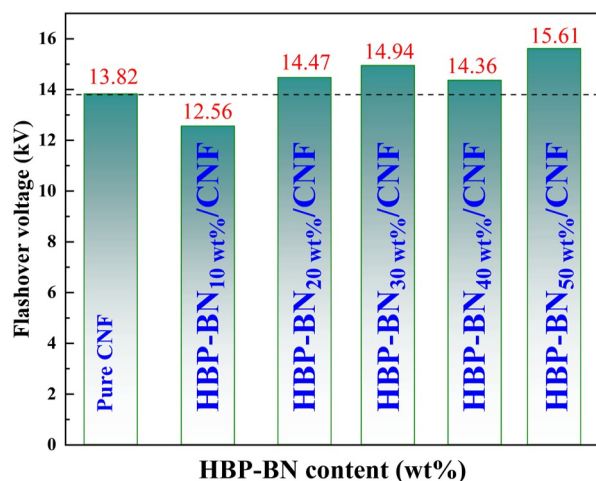


FIGURE 8 | Flashover voltage of pure CNF and HBP-BN/CNF composite films.

approaches the percolation threshold, high-density BN dispersion amplifies interfacial polarisation, slightly elevating shallow trap energy. However, the stabilisation of deep trap energy reflects inherent limitations imposed by the wide bandgap properties of BN, which restrict further energy-level elevation. Although the charge dissipation rate is moderated by the elevated energy level of shallow traps (0.761–0.802 eV), the predominance of deep traps (0.845–0.861 eV) ensures sustained bulk charge stabilisation. This synergistic trapping mechanism critically underpins the material's exceptional dielectric insulation under high-field conditions.

The alterations in shallow trap and deep trap energies induced by HBP-BN incorporation critically modulate charge transport and storage dynamics in the composite films. Shallow traps transiently localise carriers, reducing hopping barriers and enhancing charge mobility along CNF pathways, as evidenced by the dielectric stability ($\tan\delta < 0.016$) under operational fields. Conversely, deep traps immobilise charges at HBP-BN/CNF interfaces through high-strength hydrogen bonding and π - π stacking, suppressing ion migration and space charge accumulation. This dual-trap configuration establishes an energy-selective charge reservoir: shallow traps facilitate rapid charge transfer during dielectric polarisation, while deep traps prolong charge retention by elevating the energy barrier for de-trapping (> 0.845 eV). Regarding long-term electrical performance, this dual-trap configuration critically governs charge dynamics: shallow traps facilitate rapid surface charge dissipation to mitigate transient field distortion, whereas deep traps immobilise bulk charges via physical barrier effects of BN nanosheets, suppressing space charge accumulation. This synergistic charge-trapping mechanism ensures sustained dielectric stability through complementary deep and shallow trap interactions. The wide bandgap of BN fundamentally governs deep trap energy levels, providing persistent charge immobilisation, while simultaneously permitting tunable shallow trap energy for controlled surface dissipation. The system thus achieves an optimal equilibrium between rapid charge neutralisation and long-term bulk stabilisation, as demonstrated by excellent volume resistivity retention and minimal flashover voltage variation under operational conditions.

In summary, the SPD result elucidates distinct regulatory mechanisms for shallow and deep traps: shallow trap energy fluctuates via dynamic interfacial polarisation and dispersion states, whereas deep trap energy remains constrained by the intrinsic properties of BN, showing only minor enhancement through physical barrier effects. This synergistic interplay balances rapid surface charge dissipation (governed by shallow traps) and sustained bulk charge stabilisation (dominated by deep traps), offering critical insights for designing advanced insulating materials.

3.6 | Surface Flashover Characteristics of HBP-BN/CNF Films

As shown in Figure 8, the surface flashover voltage of HBP-BN/CNF composite films exhibits a distinct non-monotonic dependence on filler content, decreasing from 13.82 kV for pure CNF to 12.56 kV at 10 wt% HBP-BN, followed by a gradual increase to 14.47 kV (20 wt% HBP-BN), peaking at 14.94 kV (30 wt% HBP-BN) and reaching a maximum of 15.61 kV at 50 wt% HBP-BN after a transient decline to 14.36 kV at 40 wt%. This behaviour arises from the dynamic interplay between HBP-BN dispersion states, interfacial polarisation intensity and the competitive formation of insulating networks [43]. At 10 wt% HBP-BN, inhomogeneous dispersion of HBP-BN nanosheets generates interfacial micro holes and weakly bonded regions, which act as electric field distortion “hotspots,” accelerating pre-discharge initiation. Concurrently, partial coverage of the native surface traps of CNF by HBP reduces charge trapping efficiency, collectively contributing to voltage reduction. As the HBP-BN content increases to 20 wt%, the physical barrier effect of BN dominates, where its wide bandgap suppresses carrier migration and prolongs surface discharge paths, facilitating a recovery in flashover voltage to 14.47 kV. The peak flashover voltage of 14.94 kV for 30 wt% HBP-BN/CNF film corresponds to a critical interfacial optimisation state. SEM image (Figure 5d) confirms uniform dispersion of HBP-BN nanosheets within the CNF matrix, while HBP chains form a dense interfacial layer. This configuration reduces shallow trap energy to 0.784 eV (Figure 7f), enhances surface charge dissipation and elevates deep trap energy to 0.861 eV, enabling bulk charge immobilisation that increases volume resistivity by over an order of magnitude (Figure 5i). Maxwell–Wagner–Sillars polarisation at BN-CNF heterointerfaces further redistributes localised electric fields, suppressing charge accumulation.

At 40 wt% HBP-BN, proximity to the percolation threshold induces complex interfacial charge migration pathways because of high-density BN dispersion, transiently reducing shallow trap dissipation efficiency and lowering the voltage to 14.36 kV. For 50 wt% HBP-BN, a continuous BN insulating network obstructs long-range charge transport, stabilises deep traps at 0.858 eV (Figure 7h) and optimises electric field homogeneity through intensified interfacial polarisation, collectively driving the flashover voltage to 15.61 kV. Enhanced thermal stability (55.36% residual mass at 800°C, Figure 5j) further corroborates the critical role of the barrier effect of BN in coordinating electrical and thermal performance.

In summary, this study demonstrates that HBP modification of BN enables performance breakthroughs at high filler loadings by optimising interfacial charge transport pathways. The non-monotonic voltage evolution reflects the transition from defect-dominated interfaces to percolation networks, governed by the competition between shallow and deep trap charge dissipation mechanisms. These findings highlight the decisive influence of interfacial engineering in advancing the dielectric and insulation properties of polymer-ceramic composites.

4 | Conclusions

This study developed a dielectric insulation composite film through HBP-mediated interfacial functionalisation of BN and vacuum-assisted self-assembly of 1D CNF with 2D HBP-BN. The composite demonstrates exceptional comprehensive performance: flexible bendability (adapting to complex deformations), high thermal stability (55.36% residual mass at 800°C), optimised dielectric properties (dielectric constant $\epsilon = 5.8$ at 10 MHz, dielectric loss $\tan \delta < 0.016$) originating from dynamic regulation of shallow trap energy levels (0.761–0.802 eV) and stabilised deep trap energy distribution (0.845–0.861 eV), as well as significantly enhanced insulation performance (volume resistivity increased by > 1 order of magnitude, surface flashover voltage elevated from 13.82 to 15.61 kV). The performance improvements are primarily attributed to two mechanisms: (1) The three-dimensional branched topology of HBP enables in situ crosslinking between its terminal active groups and CNF hydroxyls, effectively suppressing BN nanosheet agglomeration and optimising interfacial compatibility. (2) The synergistic interplay between the templating effect of 1D CNF and the barrier function of 2D HBP-BN constructs a 3D charge-blocking network, achieving uniform electric field distribution and controlled carrier migration pathways. Compared to traditional polyethylene (PE) and epoxy resins (EP), the HBP-BN/CNF composite films exhibit superior integrated dielectric-insulation performance: delivering ultra-low dielectric loss ($\tan \delta < 0.016$ vs. EP: 0.02–0.04) and higher flashover voltage (15.61 kV vs. EP: < 12 kV). Additionally, the thermal stability surpasses the degradation threshold of PE ($> 80^\circ\text{C}$) and the glass transition of EP ($\sim 150^\circ\text{C}$), coupled with mechanical flexibility unattainable in brittle EP systems. While the composite delivers exceptional dielectric insulation performance, key limitations persist: interfacial degradation under prolonged thermoelectrical stress reduces volume resistivity, and trap energy drift under dynamic mechanical fatigue remains unquantified, both warranting dedicated investigation in future research. In conclusion, this work confirms that hyperbranched interfacial engineering overcomes the traditional trade-off between filler dispersion and dielectric-insulation performance, providing a promising methodology for developing BN-based functional materials with potential applications in high-voltage insulation systems for advanced electronic and electrical equipment.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available because of privacy or ethical restrictions.

References

1. A. Bjurström, H. Edin, H. Hillborg, et al., “A Review of Polyolefin-Insulation Materials in High Voltage Transmission; From Electronic Structures to Final Products,” *Advanced Materials* 36, no. 52 (2024): 2401464.
2. J. Zhou, S. Wang, J. Zhang, et al., “Enhancing Bioinspired Aramid Nanofiber Networks by Interfacial Hydrogen Bonds for Multiprotection Under an Extreme Environment,” *ACS Nano* 17, no. 4 (2023): 3620–3631.
3. X. Wang, W. Cao, Z. Su, et al., “Fabrication of High Thermal Conductivity Nanodiamond/Aramid Nanofiber Composite Films With Superior Multifunctional Properties,” *ACS Applied Materials & Interfaces* 15, no. 22 (2023): 27130–27143.
4. J. W. Zha, M. S. Zheng, B. H. Fan, and Z. M. Dang, “Polymer-Based Dielectrics With High Permittivity for Electric Energy Storage: A Review,” *Nano Energy* 89 (2021): 89106438.
5. Z. You, L. Weng, L. Guan, et al., “Preparation of Magnetic-Oriented Electronic Packaging Composite Materials With Improved Thermal Conductivity and Insulating Properties by Filling Magnetic BN@Fe₃O₄ Core-Shell Particles Into Epoxy,” *High Voltage* 10, no. 1 (2025): 219–227.
6. A. Bashir, M. Maqbool, U. Ali, et al., “Emerging Trends and Challenges in Thermal Interface Materials: A Comprehensive Perspective From Fundamentals to Applications,” *Materials Science and Engineering: R: Reports* (2025): 164100968.
7. C. Lee, Y. Noh, H. Cho, and J. Lee, “Fabrication of Insulation Spacers for Ultra-High Voltage Gas-Insulated Switchgear Using Eco-Friendly Bio-Based Epoxy Composite Material Characterisation Evaluation,” *High Voltage* 10, no. 2 (2025): 451–457.
8. V. Guerra and C. W.T. McNally, “Thermal Conductivity of 2D Nano-Structured Boron Nitride (BN) and Its Composites With Polymers,” *Progress in Materials Science* 100 (2019): 170–186.
9. J. Cao, T. L. Meng, X. Zhang, C. K. I. Tan, A. Suwardi, and H. Liu, “On Functional Boron Nitride: Electronic Structures and Thermal Properties,” *Materials Today Electronics* 2 (2022): 2100005.
10. Y. Yang, Y. Gao, H. Peng, et al., “High-Oriented Two-Dimensional BN Reinforced Composites With High Thermal Conductivity via Simple Vertical Method,” *Ceramics International* 51, no. 19 (2025): 29055–29063.
11. D. Hu, H. Liu, Y. Guo, M. Yang, and W. Ma, “Interfacial Design of Nanocellulose/Boron Nitride Nanosheets Composites via Calcium Ion Cross-Linking for Enhanced Thermal Conductivity and Mechanical Robustness,” *Composites Part A: Applied Science and Manufacturing* 158 (2022): 158106970.
12. J. Lin, J. Dong, B. Chen, et al., “Controllable Exfoliation of Hexagonal Boron Nitride and Tailored Three-Dimensional Network for Highly Thermally Conductive Polymer Composites,” *Journal of Applied Polymer Science* 141, no. 10 (2024): e55094.
13. A. Bashir, M. Maqbool, R. Lv, et al., “Surface Modified Boron Nitride Towards Enhanced Thermal and Mechanical Performance of Thermoplastic Polyurethane Composite,” *Composites Part B: Engineering* 218 (2021): 218108871.
14. C. Sun, J. Zhao, D. Zhang, H. Guo, X. Wang, and H. Hu, “Covalent Functionalization of Boron Nitride Nanosheets via Reductive Activation,” *Nanoscale* 12, no. 35 (2020): 18379–18389.
15. R. C. Zhang, D. Sun, A. Lu, et al., “Microplasma Processed Ultrathin Boron Nitride Nanosheets for Polymer Nanocomposites With Enhanced

- Thermal Transport Performance," *ACS Applied Materials & Interfaces* 8, no. 21 (2016): 13567–13572.
16. P. Amir, B. Yoshio, and D. Golberg, "Plasma-Assisted Interface Engineering of Boron Nitride Nanostructure Films," *ACS Nano* 8, no. 10 (2014): 10631–10639.
 17. K. Wu, J. Fang, J. Ma, et al., "Achieving a Collapsible, Strong, and Highly Thermally Conductive Film Based on Oriented Functionalized Boron Nitride Nanosheets and Cellulose Nanofiber," *ACS Applied Materials & Interfaces* 9, no. 35 (2017): 30035–30045.
 18. J. Yu, X. Huang, C. Wu, X. Wu, G. Wang, and P. Jiang, "Interfacial Modification of Boron Nitride Nanoplatelets for Epoxy Composites With Improved Thermal Properties," *Polymer* 53, no. 2 (2012): 471–480.
 19. B. V. Tawade, M. Singh, I. E. Apta, et al., "Polymer-Grafted Nanoparticles With Variable Grafting Densities for High Energy Density Polymeric Nanocomposite Dielectric Capacitors," *JACS Au* 3, no. 5 (2023): 1365–1375.
 20. T. Aoki and D. Aoki, "Topology-Transformable Polymers: Linear-Branched Polymer Structural Transformation via the Mechanical Linking of Polymer Chains," *Polymer Journal* 50, no. 1 (2018): 127–147.
 21. Y. Zhao, Z. Wu, R. Huang, L. Li, and G. Ma, "Hyperbranched Boron Nitride Structure-Based Epoxy Composite: Simultaneous Enhancement of Mechanical Properties, Thermal Conductivity, and Superior Electrical Insulation at Cryogenic Temperatures," *Polymer Composites* 45, no. 9 (2024): 7967–7978.
 22. Y. Zhao, Z. Wu, S. Guo, et al., "Hyperbranched Graphene Oxide Structure-Based Epoxy Nanocomposite With Simultaneous Enhanced Mechanical Properties, Thermal Conductivity, and Superior Electrical Insulation," *Composites Science and Technology* 217 (2022): 217109082.
 23. X. Liang, X. Li, Y. Tang, X. Zhang, W. Wei, and X. Liu, "Hyperbranched Epoxy Resin-Grafted Graphene Oxide for Efficient and All-Purpose Epoxy Resin Modification," *Journal of Colloid and Interface Science* 611 (2022): 105–117.
 24. Z. Lu, H. Zhang, M. Toivakka, and C. Xu, "Current Progress in Functionalization of Cellulose Nanofibers (CNFs) for Active Food Packaging," *International Journal of Biological Macromolecules* 267 (2024): 267131490.
 25. X. Yang, M. S. Reid, P. Olsén, and L. A. Berglund, "Eco-Friendly Cellulose Nanofibrils Designed by Nature: Effects From Preserving Native State," *ACS Nano* 14, no. 1 (2020): 724–735.
 26. D. Zhang, X. Feng, J. Jiang, T. Zhang, and Z. Yang, "Preparation Method and Performance Study of Nanocellulose/Epoxy Resin Composite Materials Modified by Multiple Approaches," *High Voltage* 10, no. 5 (2025): 1083–1098.
 27. A. Dufresne, "Nanocellulose: A New Ageless Bionanomaterial," *Materials Today* 16, no. 6 (2013): 220–227.
 28. B. Liu, J. Zeng, P. Li, et al., "Flexible Nanocellulose-Based Layered Films by Crosslinking Phosphorus Lignin Nanoparticles and Functionalized Boron Nitride Nanosheets for Flame-Resistant and Thermal Conductivity Applications," *Journal of Materials Chemistry A* 11, no. 44 (2023): 24057–24071.
 29. C. Gao, and D. Yan, "Hyperbranched Polymers: From Synthesis to Applications," *Progress in Polymer Science* 29, no. 3 (2004): 183–275.
 30. Y. Gao, J. Li, B. Xu, B. Ye, and B. Du, "Effect of Chemical Corrosion on Charge Transport Behaviour in Epoxy/Al₂O₃ Nanocomposite Irradiated by Gamma Ray," *High Voltage* 7, no. 1 (2022): 52–63.
 31. S. Shaybanizadeh, and A. N. Chermahini, "Fabricating Boron Nitride Nanosheets From Hexagonal BN in Water Solution by a Combined Sonication and Thermal-Assisted Hydrolysis Method," *Ceramics International* 47, no. 8 (2021): 11122–11128.
 32. J. Xia, X. Ba, Q. Zhang, et al., "Thermal Degradation Kinetics of Carbon Black Grafted With Hyperbranched Poly(Amine-Ester)s," *Journal of Gene Medicine* 9, no. 7 (2007): 605–612.
 33. F. Yao, Z. Jia, J. Dong, et al., "Boosting the Charge Transport and Stability of Metal Halide Perovskite With Boron Nitride Nanosheets for X-Ray Detection and Imaging," *Chemical Engineering Journal* 480 (2024): 480147945.
 34. W. Wei, Y. Zhang, H. Chen, Z. Su, D. Lan, and J. Zha, "Nano-BN and Nano-Cellulose Synergistically Enhanced the Mechanical, Thermal, and Insulating Properties of Cellulose Insulating Paper," *Composites Science and Technology* 256 (2024): 256110748.
 35. J. Huang, Y. Zhou, L. Dong, Z. Zhou, and R. Liu, "Enhancement of Mechanical and Electrical Performances of Insulating Presspaper by Introduction of Nanocellulose," *Composites Science and Technology* 138 (2017): 13840–13848.
 36. W. Hou, L. Yang, F. Yin, Y. Mo, R. Liao, and Y. Yuan, "Preparation of a Novel Cellulose Insulation With Network Structure by Citric Acid Crosslinking," *IEEE Transactions on Dielectrics and Electrical Insulation* 28, no. 4 (2021): 1171–1180.
 37. Y. Mo, L. Yang, W. Hou, et al., "Preparation of Cellulose Insulating Paper of Low Dielectric Constant by OAPS Grafting," *Cellulose* 26, no. 12 (2019): 7451–7468.
 38. W. Wei, Y. Zhang, H. Chen, C. Xu, J. Zha, and S. Nie, "Molecular Dynamics Simulations Guided the Preparation of Nano-Silica/Polyimide/Cellulose Composite Insulating Paper," *Materials & Design* 233 (2023): 233112176.
 39. J. Ren, G. Jiang, Z. Wang, et al., "Highly Thermoconductive and Mechanically Robust Boron Nitride/Aramid Composite Dielectric Films From Non-Covalent Interfacial Engineering," *Advanced Composites and Hybrid Material* 7, no. 1 (2023): 5.
 40. Y. Gao, N. Li, J. Li, B. Du, and Z. Liu, "Charge Transport Behavior in Gamma-Ray Irradiated Poly(Ethylene Terephthalate) Estimated by Surface Potential Decay," *High Voltage* 6, no. 3 (2021): 435–447.
 41. Y. Gao, B. Xu, X. Wang, and T. Jia, "Charge Transport in Low Density Polyethylene Based Micro/Nano-Composite With Improved Thermal Conductivity," *Journal of Physics D: Applied Physics* 52, no. 28 (2019): 285302.
 42. V. Kumar, S. Jain, A. Kapoor, J. Poortmans, and R. Mertens, "Trap Density in Conducting Organic Semiconductors Determined From Temperature Dependence of J–V Characteristics," *Journal of Applied Physics* 94, no. 2 (2003): 1283–1285.
 43. H. C. Miller, "Flashover of Insulators in Vacuum: The Last Twenty Years," *IEEE Transactions on Dielectrics and Electrical Insulation* 22, no. 6 (2015): 3641–3657.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting Information S1: hve270114-sup-0001-suppl-data.docx.